

TYPESCRIPT CHEATSHEET

Uneori este dificil sa ne dam seama ce fel de date sunt folosite in JavaScript datorita tiparii dinamice, motiv pentru care a aparut TypeScript care permite specificarea tipului datelor si verifica tipurile la compile time, nu la run time.

<https://www.typescriptlang.org>

Download TypeScript & all you need:

<https://www.typescriptlang.org/download>

TypeScript este un superset al JavaScript, nu este un limbaj diferit.

NPM = Node Package Manager

Default package manager pentru Node.js runtime environment.

Un fisier cu cod in Typescript va avea extensia .ts

Comanda "tsc file.ts" va genera un fisier .js provenit din .ts.

```
let firstName: string = "Dylan";
```

```
let firstName = "Dylan";
```

In a doua varianta spunem ca TypeScript infera tipul variabilei.

Daca tipul nu se potriveste cu cel de la declarare, TypeScript returneaza o eroare, dar JavaScript nu.

Tipul "any" dezactiveaza type checking:

```
let v: any = true;
v = "string"; // no error
Math.round(v); // no error
```

TypeScript aduce in plus, pe langa tiparea statica, un approach mai clasic asupra paradigmei OOP din JavaScript, **introducand folosirea interfetelor:**

```
// Define an interface for a person
interface Person {
  firstName: string;
  lastName: string;
  age: number;
  email?: string; // Optional property
}
```

```
// Create an object that adheres to the Person interface
const johnDoe: Person = {
  firstName: "John",
  lastName: "Doe",
  age: 30,
  email: "john.doe@example.com"
};
```

si a mostenirii:

```
// Derived class (inherits from Animal)
class Dog extends Animal {
  breed: string;

  constructor(name: string, breed: string) {
    // Call the constructor of the base class using super
    super(name);
    this.breed = breed;
  }

  // Override the makeSound method
  makeSound(): void {
    console.log("Woof! Woof!");
  }
}
```

Sintaxa pentru arrays difera:

```
const names: string[] = [];
names.push("Dylan"); // no error
// names.push(3); // Error: Argument of type 'number' is not
// assignable to parameter of type 'string'.
```

sau

```
const numbers = [1, 2, 3]; // inferred to type number[]
numbers.push(4); // no error
```